Team TYS

FinTech / AI in Finance · final defence

Mode: Partner product — MKBANK, micro & family-business credit department

Partners: MKBANK and the Central Bank of Uzbekistan · October 2026

Live prototype https://tys-fintech.vercel.app **Repository** https://github.com/hmunisbekk-droid/TYS-fintech

The regulation names the evidence — the self-employed have exactly one line

CBU Regulation No. 3618 for banks, cl.19 — documented income; cl.18 — receipts on the borrower's accounts

- **Employer-confirmed salary** — no employer — the borrower is the business
- **Pension** — working-age applicant, not a pensioner
- **Funded-pension contributions** — none: contributions come through an employer
- ✓ **Receipts credited to the borrower's bank accounts (cl.18)** — the only line a self-employed borrower has
- **Tax data incl. the annual income declaration** — self-employed income is untaxed; no declaration is filed
- **Interest, dividends, rental income** — rarely applicable; easy to document when it is

50% limit

ПДН limit for bank loans (cl.3); income = mean of 6–12 months of receipts (cl.13, cl.18); 50–100% only inside a 15% quota

~140,000

projects MKBANK plans under its lending programmes in 2026 (47,975 micro-projects financed in 2025); 1,062 mahalla agents

60–90 min

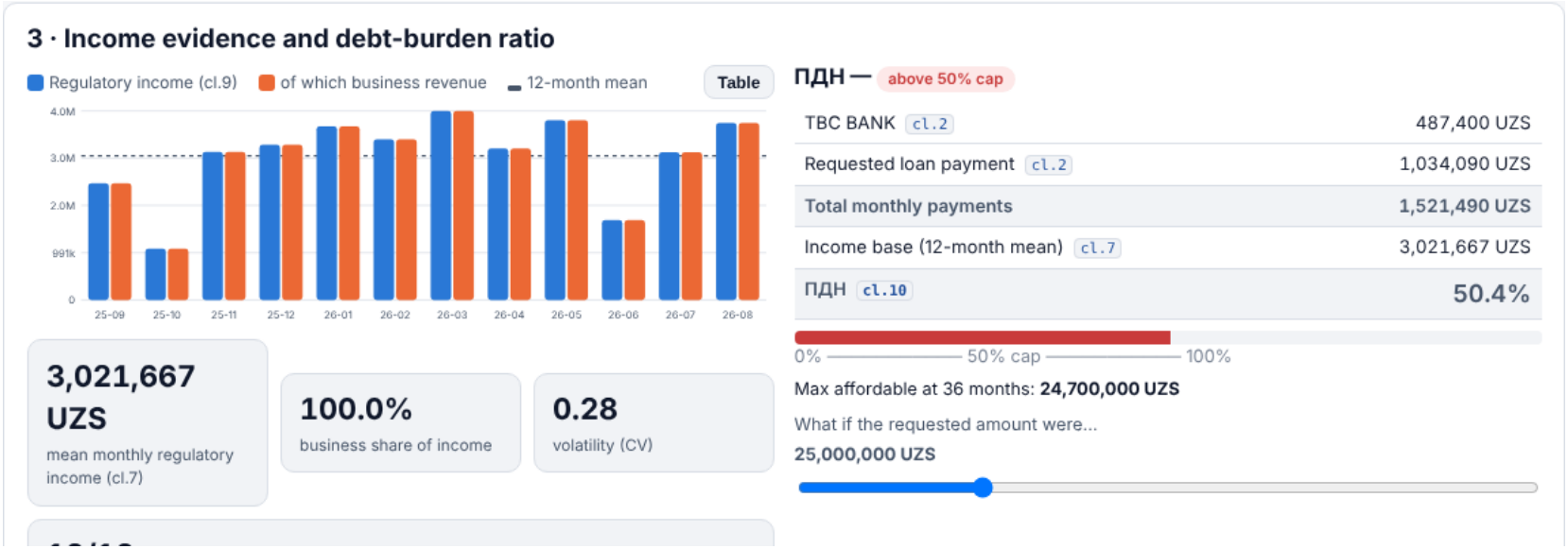
an officer's manual reading of 12 months of statements per application (our estimate; pilot measures it)

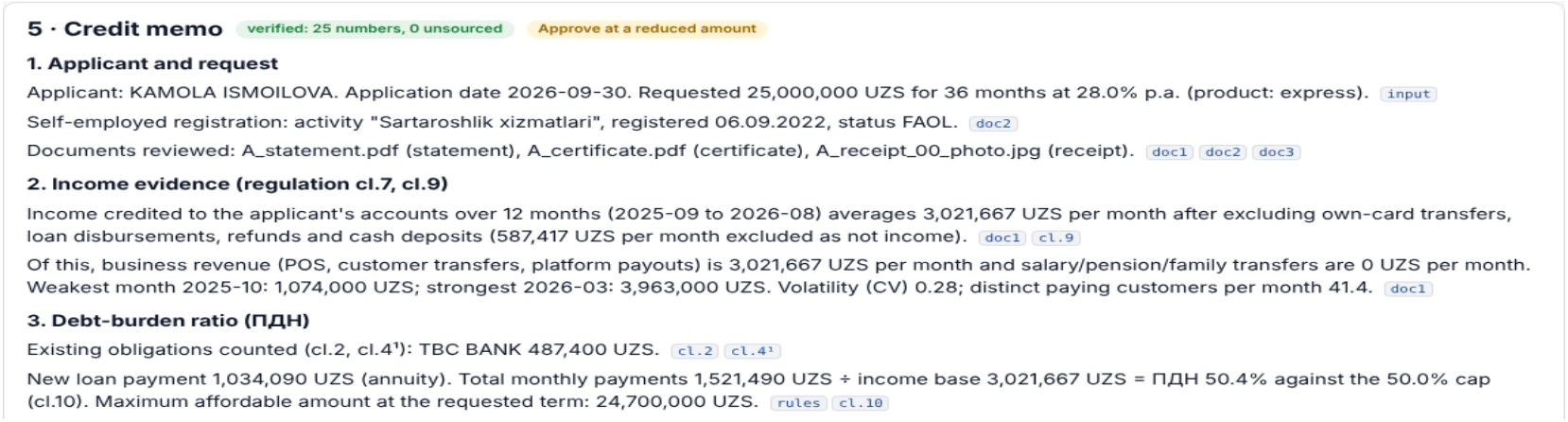
0 lines

of official income records for ~790k registered self-employed (untaxed, unfiled, no cash register required)

Sources: CBU Regulation No. 3618 (Board decision 7/3, 8 Apr 2025; lex.uz/uz/docs/7491592) · spot.uz 27 Mar 2026 (MKBANK, bank's own figures) · yuz.uz (self-employed registrations, 2021) — full list in Appendix D

What the officer gets in under five seconds





Regulatory income

cl.13 mean of 6–12 months (median option, cl.14) of account receipts (cl.18); own-card transfers, loan proceeds, refunds and cash deposits excluded

ПДН with citations

36/60/180-month spreading, guarantor rule, cl.4 exemptions, 50% limit, the 50–100% quota band and the new-self-employed quota (cl.3), maximum affordable amount

Integrity report

balance chain, duplicates, coverage, gaps, pre-application spikes, circular transfers, holder match

Verified memo

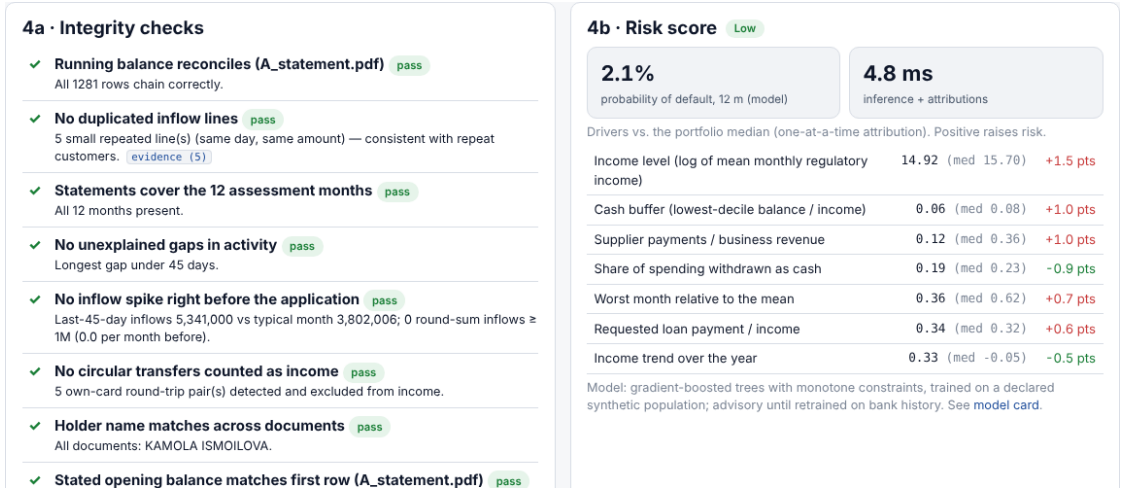
every number traces to its source — statement lines (click to open them), a clause, or the model; the verifier refuses unsourced numbers; hash-chained audit log

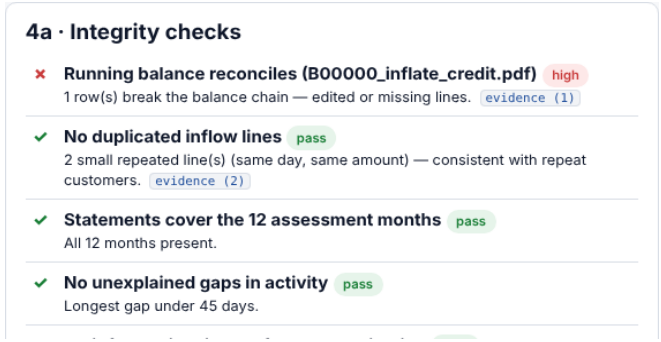
Live demo — on our samples and on anything you upload

Open on your laptop or phone

https://tys-fintech.vercel.app

1. Load sample A — statement PDF, Soliq certificate, receipt photo parse in the tab (no server, no API)
2. Income series, ПДН with clause citations, maximum affordable amount; drag the slider to cross the 50% limit
3. Integrity checks and risk score with drivers; click any evidence link to see the statement line
4. Memo: verifier badge shows how many numbers were checked and that none is unsourced
5. Record the decision; export the case with its hash-chained audit log
6. Then: Load tampered statement — the balance chain breaks at the inflated inflow; upload your own documents





Scan to open the live prototype on your phone (New case → Load sample A)

Seven components, one optional language model, zero API keys

1 · Intake

pdf.js with coordinates; XLSX/CSV; tesseract.js OCR uzb+rus+eng, self-hosted; field confidence, tap-to-confirm

2 · Categoriser

multilingual rules + counterparty patterns (recurring payer → family; same mask both ways → own card)

3 · Cash-flow

12-month series; 18 features identical in TypeScript and Python (parity test)

4 · Rule engine

ПДН clause by clause (Regulation 3618): 36/60/180-month spreading, guarantor rule, cl.4 exemptions, 50% limit, 15% quota bands, back-solved amount

5 · Risk model

XGBoost, monotone constraints, JSON trees evaluated in the browser; attributions vs portfolio median

6 · Integrity

balance chain, duplicates, coverage, gaps, round-sum spikes, circular transfers, holder match

7 · Memo + audit

template from a fact table; verifier rejects unsourced numbers; SHA-256 hash-chained log

Perimeter

Prototype: static site, every byte of a document stays in the browser. Production: the same bundle on the bank's server, SSO, audit log in the bank DB, core-banking pull for own customers. LLM optional, narrative only, behind the same verifier.

Speed: statement 1.2–1.8 s · receipt OCR 1.6 s · everything else < 50 ms · marginal cost per application: zero

Human in the loop: the engine recommends and computes the maximum affordable amount; the officer decides; an override needs a reason, written into the chain.

Measured through the real app, not typed into a slide

60/60

generator PDFs: exact rows, balance chain reconciled (2 layouts, RU/UZ)

8/8

unseen layouts exact — CSV/XLSX/TSV variants incl. a wallet export (list on the model card)

40/40

tampered statements caught, 4 tamper types; 0 false alarms on clean ones

95%

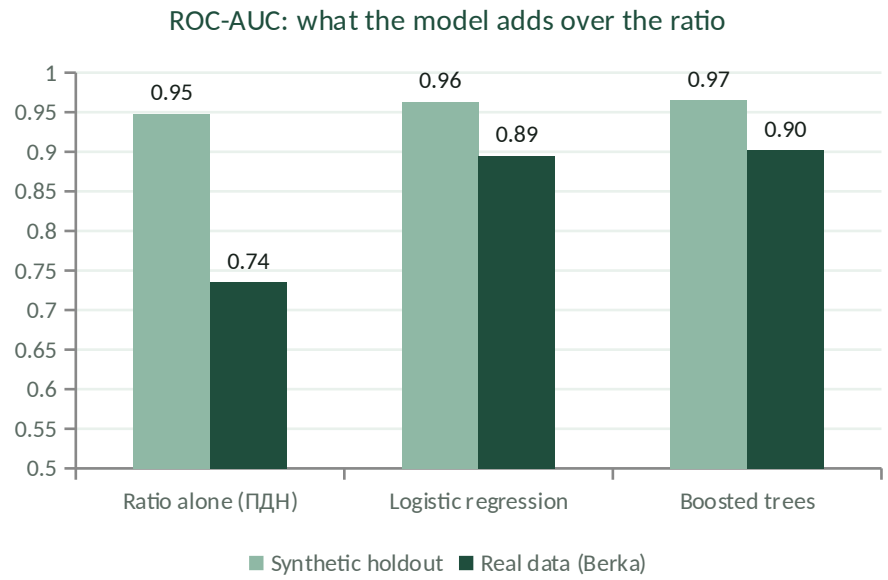
receipt OCR total exact, 60 camera-augmented photos; INN 100%, date 97%

0.965

ROC-AUC, synthetic holdout n=500; logistic 0.963, ratio alone 0.948

0.902

ROC-AUC, real bank data (Berka, 443 loans, 44 bad; 95% CI ≈ 0.86–0.94); ratio alone 0.735



Honest reading: on synthetic data the ratio already carries most of the signal by construction; on real statements the cash-flow features add +0.17 AUC over the ratio, and logistic regression comes close to the trees — we ship the trees for the monotone constraints and non-linear thresholds, not for the last 0.01. Categoriser: 99.96% on 10,481 lines of the generator's own vocabulary — will fall on real data. Memo verifier: 0 unsourced numbers across a 300-scenario sweep (amounts × terms × products × borrowers), enforced at run time.



All numbers are read from JSON produced by ml/train.py, ml/berka_validate.py, npm test and tests/*_eval.py (see the Model card page).

Caveat on the slide, not in the footnote: the risk score is advisory until retrained on the bank's own history inside the perimeter — that is the pilot's first deliverable.

No partner data, so we built a declared generator

1

Archetypes

10 trades (tailor, taxi, dehkan farmer, grocer, reseller, tutor…) with declared seasonality, customer counts, channel mix, expense structure

2

Simulation

2,000 borrowers · 5,689,589 transactions · existing loans with annuity schedules · shocks · default label from a latent affordability score (10.5% rate)

3

Rendering

60 statement sets as PDF (text layer) + XLSX + CSV in RU and UZ layouts; 180 fiscal receipts; Soliq-style certificates with QR

4

Camera model

perspective, rotation, lighting gradient, blur, sensor noise, JPEG — then measured through the same OCR the officer uses

5

Tampering

40 statements: inflated inflow, deleted debit, duplicated row, window dressing — labelled for the integrity evaluation



HUMO CARD STATEMENT
Выписка по карте

Владелец: KAMOLA ISMOILOV
Карта: 5614 **** **** 4369
Период: 2025-09-01 — 2026-08-31
Валюта: UZS

Входящий остаток: **668 582.52**

Дата	Время	Описание операции	Расход	Приход	Остаток
2025-09-01	07:33	P2P HUMO 8600 **** 9613		50 000.00	718 582.52
2025-09-01	08:33	IJARA TO'LOVI P2P 8600 **** 2210	674 000.00		44 582.52
2025-09-01	08:54	NAQD PUL KIRITISH BANKOMAT		350 000.00	394 582.52
2025-09-01	09:03	HUDUDGAZTAMINOT KOMMUNAL	131 000.00		263 582.52
2025-09-01	14:57	HUDUDIY ELEKTR TARMOQLARI KOMMUNAL	90 000.00		173 582.52
2025-09-01	16:24	UCELL ABONENT	62 000.00		111 582.52
2025-09-02	14:09	BARAKA MARKET TOSHKENT	103 000.00		8 582.52
2025-09-03	16:36	POS KAMOLA SARTAROSHLIK TERMINAL		52 000.00	60 582.52
2025-09-04	07:29	P2P HUMO 9860 **** 9719		17 000.00	77 582.52
2025-09-04	12:45	POS KAMOLA SARTAROSHLIK TERMINAL		40 000.00	117 582.52
2025-09-04	13:46	POS KAMOLA SARTAROSHLIK TERMINAL		22 000.00	139 582.52
2025-09-04	16:24	P2P HUMO 8600 **** 6177		38 000.00	177 582.52
2025-09-04	16:48	P2P HUMO 9860 **** 4617		39 000.00	216 582.52
2025-09-04	17:40	POS KAMOLA SARTAROSHLIK TERMINAL		27 000.00	243 582.52
2025-09-04	22:21	ANGLESEY TOSHKENT	63 000.00		180 582.52
2025-09-05	20:06	POS KAMOLA SARTAROSHLIK TERMINAL		32 000.00	212 582.52
2025-09-05	20:29	P2P HUMO 8600 **** 9240		31 000.00	243 582.52
2025-09-05	22:57	OPTOM SAVDO MCHJ TO'LOV	104 000.00		139 582.52
2025-09-06	14:53	POS KAMOLA SARTAROSHLIK TERMINAL		64 000.00	203 582.52
2025-09-06	17:24	HAVAS TOSHKENT	131 000.00		72 582.52
2025-09-07	11:40	P2P HUMO 9860 **** 3683		66 000.00	138 582.52
2025-09-07	13:08	POS KAMOLA SARTAROSHLIK TERMINAL		31 000.00	169 582.52
2025-09-07	20:29	OPTOM SAVDO MCHJ TO'LOV	88 000.00		81 582.52
2025-09-08	07:44	POS KAMOLA SARTAROSHLIK TERMINAL		43 000.00	124 582.52
2025-09-08	15:53	POS KAMOLA SARTAROSHLIK TERMINAL		39 000.00	163 582.52
2025-09-08	16:34	OPTOM SAVDO MCHJ TO'LOV	106 000.00		57 582.52
2025-09-08	18:41	KORZINKA TOSHKENT	42 000.00		15 582.52
2025-09-09	08:34	POS KAMOLA SARTAROSHLIK TERMINAL		50 000.00	65 582.52
2025-09-09	09:50	POS KAMOLA SARTAROSHLIK TERMINAL		27 000.00	92 582.52
2025-09-09	21:57	HAVAS TOSHKENT	50 000.00		42 582.52
2025-09-10	12:30	POS KAMOLA SARTAROSHLIK TERMINAL		30 000.00	72 582.52
2025-09-10	18:43	POS KAMOLA SARTAROSHLIK TERMINAL		46 000.00	118 582.52
2025-09-10	22:08	P2P HUMO 8600 **** 3739		34 000.00	152 582.52
2025-09-11	10:18	P2P HUMO 9860 **** 5631		33 000.00	185 582.52
2025-09-11	17:02	POS KAMOLA SARTAROSHLIK TERMINAL		37 000.00	222 582.52
2025-09-11	18:42	P2P HUMO 8600 **** 7135		44 000.00	266 582.52
2025-09-11	21:26	P2P HUMO 9860 **** 6338		30 000.00	296 582.52
2025-09-11	22:12	P2P HUMO 9860 **** 6674		34 000.00	330 582.52
2025-09-12	15:07	POS KAMOLA SARTAROSHLIK TERMINAL		27 000.00	357 582.52
2025-09-12	18:20	P2P HUMO 9860 **** 2803		24 000.00	381 582.52



KAMOLA SARTAROSHLIK
Xorazm, Mustaqillik ko'chasi 9
STIR/INN: 342056057
FISKAL CHEK № 58332
01.03.2026 13:36
Kamola
Sana
Kassir
BOSH YUVISH 45 000
3 x 15 000
SOCH OLISH 40 000
1 x 40 000
JAMI 85 000
To'landi (UZCARD) 85 000
QQS 12% 9 107
FM: UZ118914450618 FP: 135878439006
ofd.soliq.uz
Xaridingiz uchun rahmat!

Synthetic statement (Humo-style layout, Russian labels) and a camera-augmented synthetic fiscal receipt. No real person is represented; names come from a short public list, card numbers are random and masked.

```
Regenerate:  python -m synth.cli --n 2000 --render 60 --
receipts 180 --tamper 40 --seed 42
```

Also: methodology check on real anonymised bank data (Berka, PKDD'99) — not redistributed, downloaded by script.

Pilot: three months, five branches, fifty mahalla agents

Cost to build and run the pilot (assumptions, UZS)

Line	Assumption	3-month cost
Team: 3 engineers/analysts (TYS)	3 × 3 months × 15M	135M
Bank credit analyst, part-time	0.5 FTE × 3 months × 15M	22.5M
Infrastructure	1 VM in the bank DC, no GPU	≈ 3M
Integration (core banking, bureau)	bank IT, 20 person-days	≈ 30M
Data, licences, API spend	none — open-source, no external API	0
Total pilot (≈ $15k)		**≈ 190M**

Running cost per application: zero marginal (static bundle; no API, no GPU). Optional narrative LLM on one on-prem GPU: well under 1 US cent per application.

Timeline: month 1 shadow mode (engine runs beside the manual process; we measure minutes and agreement) · months 2–3 assisted mode (officer works from the memo) · success metrics: minutes per application, agreement with committee, override rate, ПДН audit findings, retrained model AUC on bank history.

Value — the officer-hours formula (every term is a slider)

90 → 25 min

document handling per application, before → with Daromad (assumption 1; the pilot measures it)

≈ 42,000 / yr

addressable applications: 30% of MKBANK's 140,000 micro-projects (bank's own 2026 target) — those outside the cl.4-exempt family programme (assumption 2)

≈ 45,000 h

officer-hours saved per year = 42,000 × 65 min ≈ 23 FTE-years at 2,000 h

≈ 4.1 bn UZS/yr

23 FTE × 15M/month × 12 — officer time only; at a 10% addressable share ≈ 1.4 bn

Payback: the ≈ 190M pilot itself (5 of 181 branches, 3 months) does not pay back — it measures the two assumptions. After roll-out the saving of ≈ 4.1 bn/yr repays the pilot within its first month; at a 10% addressable share, within two months. Not counted: faster time-to-decision for the borrower, fewer ПДН compliance findings, better risk selection — claimed only after the pilot measures them.

Who pays: MKBANK (credit department / digital budget). Who owns: micro & family-business credit department; IT deploys; mahalla agents and loan officers use it.

Hard to copy in a weekend — and what comes next

The replication test — what the same LLM tools do not give you

- Statement parsers that learn columns from the data rows (two Uzbek layouts today, any text-layer PDF, XLSX, CSV) and reconcile the balance chain
- A categoriser lexicon in Uzbek Latin, Cyrillic and Russian with counterparty-pattern rules — and the pilot plan to grow it on bank data
- A rule engine mapped clause by clause to the live bank regulation (No. 3618), with its exemptions, quota bands and the back-solved maximum amount
- A memo verifier that makes unsourced numbers impossible — zero hallucination by construction, no LLM needed
- A declared, calibrated synthetic population with rendered documents, a camera model and labelled tampering — a test bed the bank can keep using
- Evaluation through the real app, reproducible with one command each

Next steps

- Week 1–2: scanned-statement OCR page by page; lexicon growth on anonymised bank descriptions; Uzbek/Russian memo templates
- Month 1: shadow mode in 5 branches; core-banking pull for MKBANK's own account holders; credit-bureau REST for obligations
- Month 2–3: assisted mode; retrain the risk model on bank history inside the perimeter; set risk bands with the bank's risk appetite; fairness review by region
- After the pilot: Central Bank conversation — the same engine computes ПДН uniformly for any lender, a supervisory check at zero marginal cost
- Product scope today: Ommabop 2, Express, Business Online 2 (ПДН applies) and family-programme loans (exempt from the cap, capacity still assessed)

Daromad

Income evidence for borrowers who have no payslip — regulation-native, perimeter-native, verified.

Live: https://tys-fintech.vercel.app
Repo: https://github.com/hmunisbekk-droid/TYS-fintech
Team TYS

Appendix A · ПДН rule engine, clause by clause

Clause	Rule as implemented	Where it shows
cl.2	ПДН = average monthly payments on loans (microloans) ÷ average monthly income («ўртача ойлик тўловлари миқдорининг … ўртача ойлик даромадига нисбати»)	formula panel
cl.3	Ratio ≤ 50% («50 фоиздан ошмаслиги лозим»). Up to 15% of outstanding loans may carry up to 100% (amendment 31/9, 2025) — bank-level quota → committee. Self-employed registered < 6 months: income-generating loans within 15% without the ratio	status: pass / quota / fail; cl.3 ¶4 badge
cl.4	Not applicable to loans to individual entrepreneurs, family-entrepreneurship programmes and education loans («татбиқ этилмайди»)	exemption badge
cl.7	All agreements with all lenders incl. the application under review; co-borrower loans counted; guarantor loans only when > 30 days overdue («30 кундан ортиқ муддати ўтган»)	obligations table, DPD column
cl.8	Annuity spreading: microloan > 36 months → 36; other loan > 60 → 60; mortgage > 180 → 180	per-line note
cl.12	No more than 3 working days between the calculation and the issue («3 иш кунидан ошмаслиги керак»)	case date
cl.13	Average income over a continuous 6–12-month period («камида 6 ой … 12 ойдан ошмаган»): 1/12 of the last 12 months, or 1/n for 6–11	income base, months used
cl.14	The median may replace the mean when it is lower	median switch
cl.18	Receipts credited to the borrower's and co-borrower's bank accounts count as income («банк ҳисобварақларига тушган тушумларини инобатга олади»)	statement-based income
cl.19	Documented evidence: salary, pension, funded-pension contributions, tax data incl. the annual declaration, interest/dividends/rent — net of taxes	memo section 2

CBU Board decision 7/3 of 8 Apr 2025, reg. No. 3618 (22 Apr 2025), in force 24 Jul 2025, amended 5 Dec 2025 (in force 21 Jan 2026) — lex.uz/uz/docs/7491592. Regulation 3205 (lex.uz 4654149) covers MFOs and pawnshops only since 24 Jul 2025 and is not used. Checked 11 Oct 2026; quotes in docs/REGULATION_QUOTES.md.

Appendix B · Failure cases, perimeter, human in the loop

Known failure cases (also on the Model card page)

- Scanned statements without a text layer: not OCR'd page by page yet — ask for the XLSX/CSV export
- One signed amount column and no balance column: signs taken from keywords only
- Merchant strings outside the lexicon (mixed scripts, transliteration) → default category, visible in the table
- Dark or heavily skewed receipt photos: OCR confidence < 60% → fields flagged for confirmation, never guessed
- Cash-only businesses: nothing to reconstruct — the memo says so; the officer attests
- Synthetic-to-real gap of the risk score until retrained inside the bank; risk bands are placeholders

Perimeter, audit, human in the loop

- Prototype: static site; documents are parsed in the browser and never transmitted; nothing is stored server-side
- Production: same bundle on the bank's web server (Docker, no outbound network), SSO identity, audit log and cases in the bank DB
- Own customers' statements pulled from core banking; uploads only for other banks' statements and wallets
- Every analysis stamped with rule-engine, model and categoriser versions; hash-chained SHA-256 log; export with the case
- The engine recommends; the officer decides; overrides require a reason; committee above branch limits
- Personal data processed inside the bank with recorded consent; retention per bank policy; no gender or activity-type features in the model

Appendix C · Data sources and licences

Source	Used for	Licence / terms
Daromad synthetic data set (synth/, seed 42)	training, evaluation, demo samples	MIT, this repo; generator and parameters declared; no real person
Berka / PKDD'99 Financial data set (mirror github.com/jlacko/berka-dataset)	offline methodology check	academic challenge data, no licence stated; evaluation only, not redistributed
Tesseract tessdata_fast (uzb, uzb_cyrl, rus, eng); tesseract.js	receipt OCR	Apache 2.0
pdf.js; SheetJS; Next.js; React	parsing, application	Apache 2.0 / MIT
XGBoost; scikit-learn; NumPy; pandas; OpenCV; Playwright; Jinja2; qrcode	training, rendering, evaluation	Apache 2.0 / BSD-3 / MIT
CBU Regulation No. 3618 for banks (lex.uz/uz/docs/7491592)	rule engine, citations	public legal information
stat.uz wages; CBU statistics; MKBANK public pages (sponsored article, bank's own figures)	calibration anchors, business case	public information, cited

Rules compliance: no hard-coded answers (every figure computed from uploaded documents at run time); no real personal or confidential data; all code written during the final by team TYS with LLM coding assistance (declared); deployment from the submitted repository; CI on every push.

Appendix D · References, Westminster Harvard (1/2)

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Appendix D · References, Westminster Harvard (2/2)

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